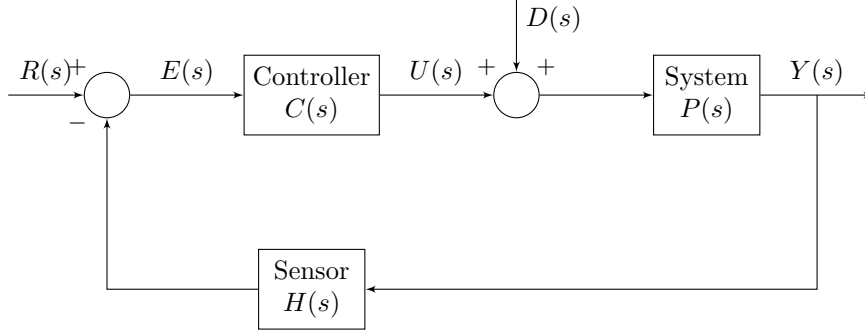


System
Disturbance
Feedback loop
open close System



1 Linear Constant Coefficient Ordinary Differential Equations (LCC-ODE)

A differential equation of the form

$$\sum_{k=0}^n a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^M b_m \frac{d^m r(t)}{dt^m},$$

where $y(t)$ is the output (response) of the system and $r(t)$ is the input to the system; and a_i and b_i are constant.

2 Laplace Transform

The Laplace transform of a function $f(t)$, defined for all real numbers $t \geq 0$, is the function $F(s)$, which is a unilateral transform defined by

$$\mathcal{L}[f](s) = F(s) = \int_{0^-}^{\infty} f(t) e^{-st} dt,$$

where $s = \sigma + i\omega$ complex frequency-domain parameter, and, 0^- is the shorthand notation for

$$\lim_{\varepsilon \rightarrow 0^+} \int_{-\varepsilon}^{\infty}.$$

2.1 Region of Convergence (RoC)

2.2 Properties of Laplace Transform

2.3 Standard Laplace Transform Pairs

Function	Time Domain $f(t)$	Laplace Domain $F(s)$	ROC
Impulse	$\delta(t)$	1	All s
Step	$u(t)$	$\frac{1}{s}$	$\Re(s) > 0$
Ramp	$tu(t)$	$\frac{1}{s^2}$	$\Re(s) > 0$
Power	$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$\Re(s) > 0$
Exponential	$e^{-at}u(t)$	$\frac{1}{s+a}$	$\Re(s) > -a$
Sine	$\sin(\omega t)u(t)$	$\frac{\omega}{s^2 + \omega^2}$	$\Re(s) > 0$
Cosine	$\cos(\omega t)u(t)$	$\frac{s}{s^2 + \omega^2}$	$\Re(s) > 0$

3 Second-Order System Analysis

Consider a linear time-invariant (LTI) second-order system described by the transfer function $G(s)$:

$$G(s) = \frac{b}{s^2 + as + b}$$

where a and b are constant coefficients. To analyze the dynamic behavior, we map these coefficients to standard physical parameters.

Definition 3.1 (Natural Frequency, ω_n): The natural frequency, ω_n , represents the frequency at which the system would oscillate in the absence of damping. It is defined as:

$$\omega_n := \sqrt{b}$$

Definition 3.2 (Damping Ratio, ζ): The damping ratio, ζ , is a dimensionless parameter that characterizes the rate at which oscillations decay. It is related to the system coefficients by:

$$\zeta := \frac{a}{2\omega_n} = \frac{a}{2\sqrt{b}}$$

Substituting these parameters back into the original expression yields the **canonical form** of the second-order transfer function:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The poles of the system, denoted as $s_{1,2}$, are the roots of the characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$. Using the quadratic formula, the poles are given by:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

4 Underdamped Second Order System

$$Y(s) = \frac{1}{s} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{A_1}{s} + \frac{A_2 s + A_3}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$Y(s) = \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)} - \frac{(\zeta/\sqrt{1 - \zeta^2})\omega_n\sqrt{1 - \zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

$$\omega_d := \omega_n\sqrt{1 - \zeta^2}$$

$$y(t) = u(t) - e^{-\zeta\omega_n t} \cos(\omega_d t)u(t) - \frac{\zeta}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t)u(t)$$

Definition 4.1 (Rise time, T_r): The time required for the waveform to go from 0.1 of the final value to 0.9 of the final value.

$$T_r = \frac{\pi - \theta}{\omega_n\sqrt{1 - \zeta^2}}, \quad \theta = \arctan\left(\frac{\sqrt{1 - \zeta^2}}{\zeta}\right)$$

Definition 4.2 (Peak time, T_p): The time required to reach the first, or maximum, peak.

$$T_p = \frac{\pi}{\omega_n\sqrt{1 - \zeta^2}}$$

Definition 4.3 (Percent overshoot, %OS): The amount that the waveform overshoots the steady-state, or final, value at the peak time, expressed as a percentage of the steady-state value.

$$\%OS = \exp\left(-\zeta\pi/\sqrt{1 - \zeta^2}\right) \times 100$$

Definition 4.4 (Settling time, T_s): The time required for the transient's damped oscillations to reach and stay within $\pm 2\%$ of the steady-state value.

$$T_s = \frac{-\ln\left(0.02\sqrt{1 - \zeta^2}\right)}{\zeta\omega_n} \approx \frac{4}{\zeta\omega_n}$$